

Big Data Project

GitHub Innovation Portal

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1. Problem Statement and Motivation

The number of open-source projects on platforms like GitHub has grown massively in recent years, making it increasingly hard to keep track of what's actually worth paying attention to. Finding promising, early-stage repositories without relying on external marketing or well-known contributors requires a lot of manual effort. As a result, developers, researchers, and tech scouts often only discover new trends through mainstream media or popular tech forums, by which point the projects are already widely known. There is currently no good automated way to surface these "hidden gems" early on. This project aims to fill that gap by building a system that tracks and visualizes emerging trends in open-source development, helping users discover innovative projects before they go mainstream.

2. Research Question and Objectives

Research Question: How can continuous GitHub event data be used to automatically detect early-stage innovation trends and identify emerging "rising star" repositories in the open-source ecosystem?

Project Objectives:

- **Data Analysis:** Analyze GitHub event streams to define metrics that indicate project momentum and early-stage innovation.
- **Visualization:** Design and implement a dashboard that visualizes technological trends and highlights emerging repositories.
- **Granular Filtering:** Add filtering options (e.g., by programming language, license type, or event type) to allow targeted exploration of specific technology areas.

3. Methodology, Data, and Implementation

The project execution is structured into three main phases. First, a robust data pipeline will be established to continuously fetch and ingest data from GitHub. Given the high frequency of user activities, as the velocity is high enough to think of the data input as a stream, followed by persistent storage in an appropriate database system. In the second phase, an exploratory data analysis (EDA) will be conducted to identify reliable indicators of emerging trends, such as sudden spikes in starring activity or specific event triggers. Finally, a prototype dashboard will be developed to visualize these metrics and allow users to filter repositories based on granular criteria.

Data Sources:

- GitHub REST API (Activity & Events endpoints, e.g., `GET /events`)
- GitHub REST API (Repository metrics, e.g., starring and forking activity)

Technical Infrastructure:

- A suitable stream processing framework (e.g., Apache Kafka or Apache Flink) for real-time data ingestion and processing
- A suitable database (e.g., NoSQL or time-series database) for storing structured event data

Expected Challenges:

- **API Rate Limiting:** Managing GitHub's request quotas efficiently while maintaining a continuous data flow.
- **Data Volume & Velocity:** Handling the high throughput of GitHub's global event stream without processing bottlenecks.
- **Data Noise:** Filtering out automated bot activities or spam repositories to isolate genuine signals of innovation.

4. Timeline and Task Allocation

Week	Phase	Task
KW 03	Setup	Project initialization, API access, Kafka pipeline configuration
KW 04	Exploration	Data stream collection, exploratory data analysis (EDA), pre-processing
KW 06	Implementation	Core system development, database integration, dashboard creation
KW 10	Evaluation	System testing, metric validation, visualization refinement
KW 13	Finalization	Final report writing, presentation preparation, project submission

Table 1: Project Timeline and Task Allocation

5. References

1. GitHub REST API Documentation. <https://docs.github.com/en/rest>